

Name: _____ Date: _____

Period: _____

Primary Objective

By the end of this lesson, I will be able to...

Vocabulary & Key Concepts

Frequency:

Relative frequency:

Frequency table:

Relative frequency table:

Distribution (of a categorical variable):

Hook

Which school has more walkers? — Fill in the missing column.

Mode	School A	Rel. Freq. A	School B	Rel. Freq. B
Bus	120	_____	30	_____
Car	80	_____	50	_____
Walk	40	_____	120	_____
Total	240	_____	200	_____

After computing relative frequencies, which school has more walkers *proportionally*?

_____ What does this tell you about using counts alone for comparison?

Building a Frequency and Relative Frequency Table

Step-by-Step Process

1. List all _____ in the first column.
2. _____ the data (use tally marks).
3. Record the _____ (count) in the second column.
4. Divide each count by _____ to get the relative frequency.
5. **Check:** relative frequencies must sum to _____.

Formula

$$\text{Relative frequency} = \frac{\text{frequency}}{\text{total}}$$

Can be expressed as a _____ (decimal)
or a _____ (multiply by 100).

Quick error check:

Sum of all relative frequencies = _____
Sum of all relative frequencies = _____%

Worked Example — Pet Ownership Survey ($n = 40$)

A sample of 40 households is surveyed about the type of pet they own most (if any). The raw data are summarized below. Complete the table.

Pet type	Tally	Frequency	Rel. Frequency
Dog		12	_____
Cat		8	_____
Fish		6	_____
Bird		4	_____
None		10	_____
Total		_____	_____

Write one AP-style sentence interpreting the most common category:

Describing a Distribution — AP Language

What to include:

- The _____ with the highest relative frequency (most common).
- The _____ with the lowest relative frequency (least common).
- The _____ of individuals in the sample (to give context).
- _____ — always include this.

Model sentence frame:

“Of the _____ [individuals] surveyed, the most common [variable] was _____ with _____% of responses. The least common was _____ with _____%.”

Practice: Using the pet table from the worked example, write a complete description of the distribution of pet type.

Guided Practice

Complete the relative frequency table. Then answer the questions.

A survey of 25 AP Statistics students asks: “How do you get to school?” Results: Car 10, Bus 8, Walk 4, Bike 3.

Mode	Frequency	Rel. Frequency
Car	10	_____
Bus	8	_____
Walk	4	_____
Bike	3	_____
Total	_____	_____

1. What proportion of students walk to school? _____
2. What percentage of students do *not* take the bus? Show your work. _____
3. Write one complete sentence describing the most common mode of transportation in context.

Connections & Big Ideas

This lesson connects forward to...

- **Lesson 1.4:** Bar charts and pie charts display the same distributions visually.
- **Unit 2:** Two-way tables extend this to two categorical variables simultaneously.
- **Unit 4:** Relative frequency $\approx$ probability when the sample is representative.

Big Idea — Write in your own words:

Why must we use relative frequency (not count) to fairly compare two groups of different sizes?

Additional Notes
